ZLabel [SimTalk]

Sets the label of the z-axis that the *Chart* designated by <Path> shows in the *Chart* window.

Remarks

Applies to the 3D chart types **3D Columns**, **3D Wire Frame**, and **3D Surface**.

Type

Attribute

Syntax

```
<Path>.ZLabel:string
```

Assignment Value

You can assign a value of data type *string*.

Example

```
MyChart.ZLabel := "Number of Stacked Rows"
```

See also

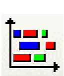
Z-Axis [text box] - Chart

GanttChart

GanttChart [object]

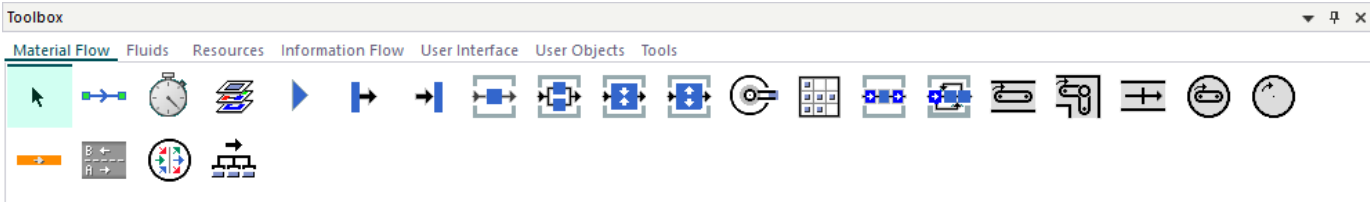
Use the object *GanttChart* for showing the chronological sequence of activities as bars on the time axis.

Image



Description

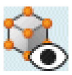
The *GanttChart* visualizes parts on resources. In this context resources are the **material flow objects**, not the **resource objects**.



In addition to the mere occupancy data, the *GanttChart* can also display failures of machines, pausing times, and blocking times.

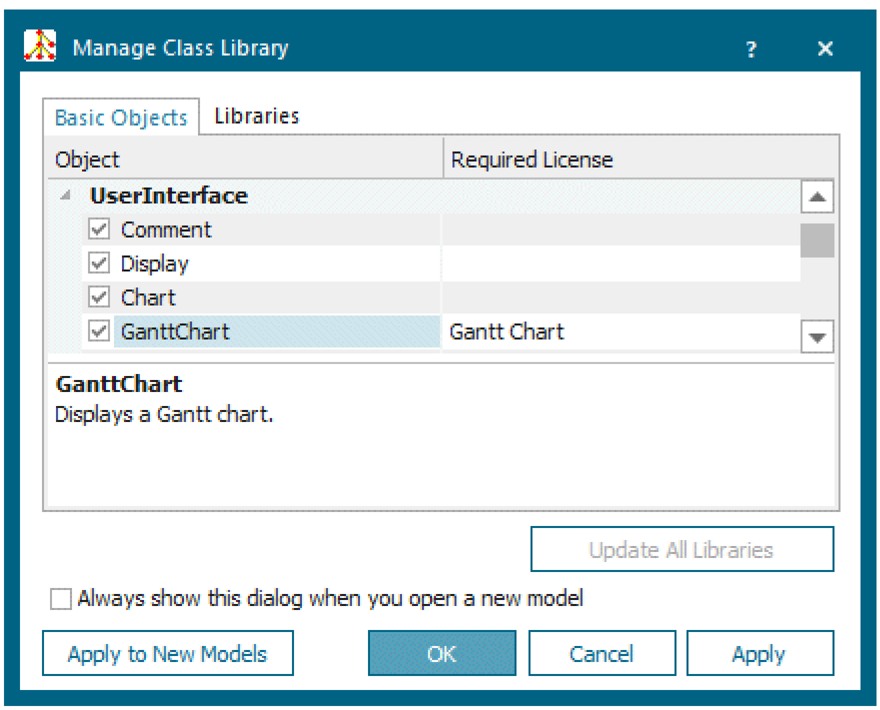
This way the *GanttChart* immediately shows which machines or stations are overloaded or have a low utilization and which production orders or products have long waiting times during the production process.

To show a **tooltip** with information about the *GanttChart*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *GanttChart*, click **Show Manipulators**  on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *GanttChart* to your simulation model, click  **Manage Class Library** > **Basic Objects > UserInterface > GanttChart** on the **Home** ribbon tab.



> **Note**
>
> The object *GanttChart* is not part of the *Plant Simulation* standard program package.

> **Note**
>
> If you are using the *GanttChart*, replace the *GanttChart* of previous versions with the *GanttChart* of the current version.
>
> Adjust the source code, which you programmed in *SimTalk* of previous versions, to the current version, compare **Methods of the GanttChart** and **Attributes of the GanttChart**.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

See also

Show Parts on Resources in a GanttChart

Dialog Box of the GanttChart

Dialog Box of the GanttChart

Double-click the icon of the *GanttChart* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under **Dialog Items of the Objects**.

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box **Edit 3D Properties**:

- Click the button **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

See also

Show Chart [button] - GanttChart

Collect Data [check box] - Chart

Collect Data [check box] - GanttChart

To make the *GanttChart* collect data during the simulation run, select this check box. To deactivate data collection, clear the check box.

Remarks

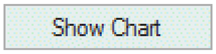
You cannot use the method *setData* and automatically collect the data simultaneously. When you select **Collect Data**, do not call the method **setData**.

SimTalk

CollectData [SimTalk] - GanttChart

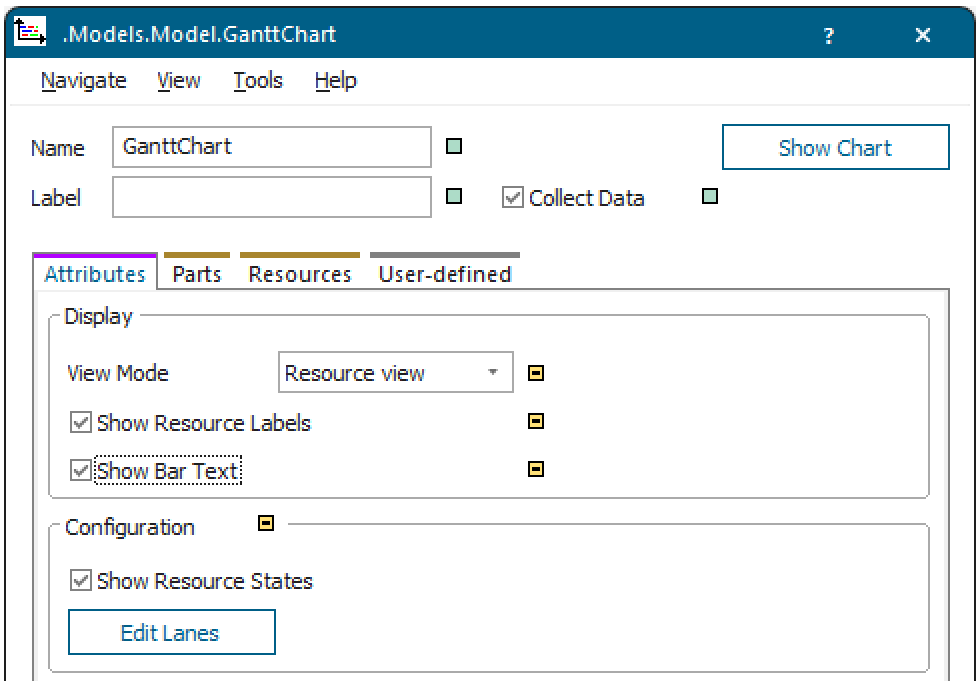
Show Chart [button] - GanttChart

To show the data, which the *GanttChart* collected in the display window, click this button.



Remarks

Define how the *GanttChart* shows the collected data on the tab **Attributes**.

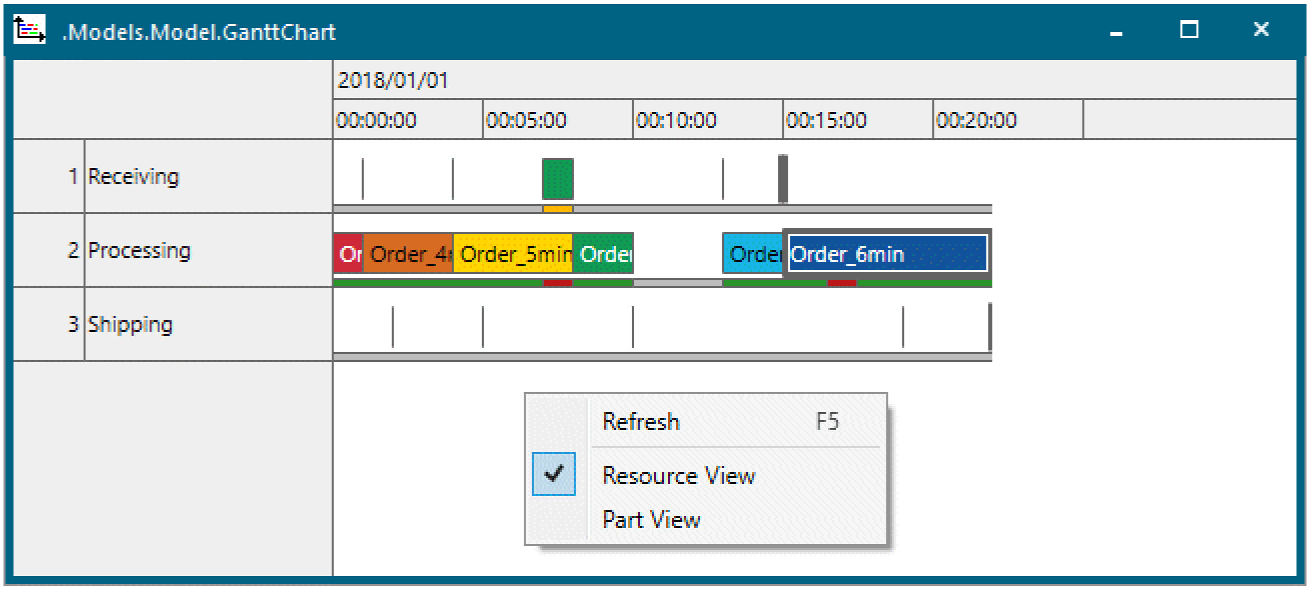


The *GanttChart* provides two views:

Resource View

Resource View shows the resources which are occupied by the parts. The *GanttChart* shows the resources along the vertical axis and the elapsing time along the horizontal axis.

> **Note**
>
> The *GanttChart* shows the resources in the order in which they processed parts for the first time. The resource, which processed a part first, is shown in the first lane in the chart. The resource, which processed a part after that, is shown in the lane below, etc.
>
> You cannot change the order of the resources.

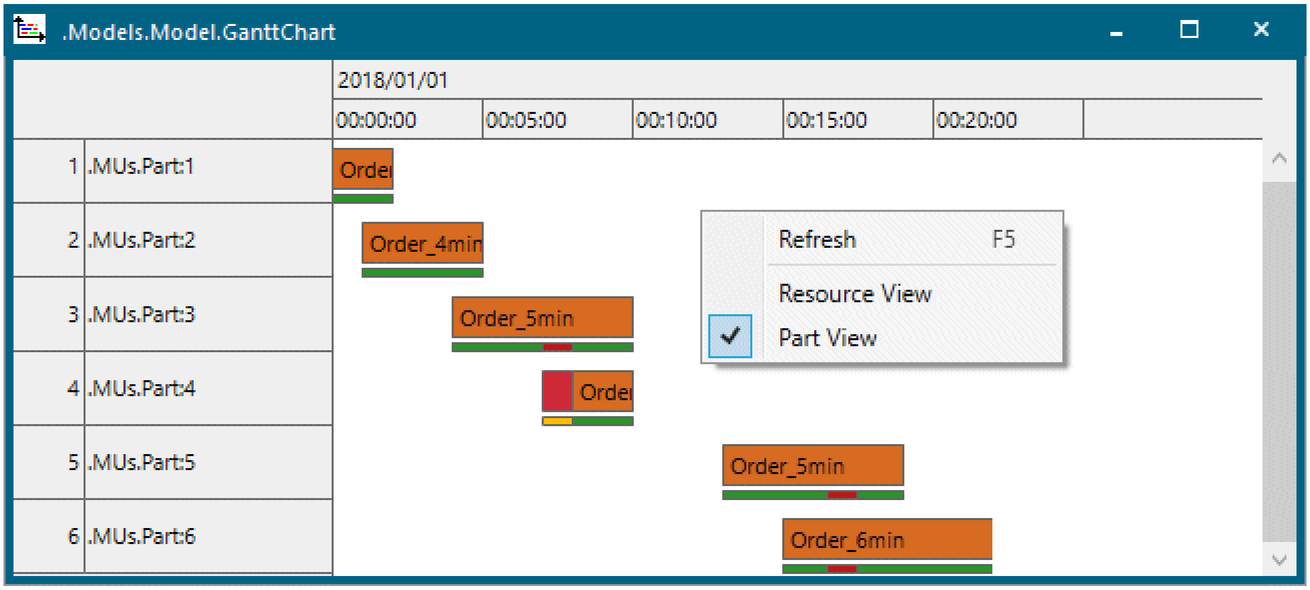


With the default settings the *GanttChart* shows a part on all resources with the same color.

Part View

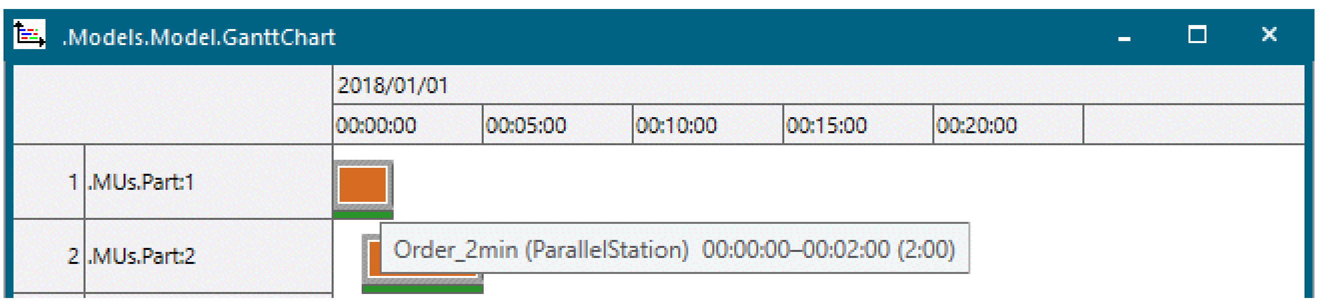
Part View shows the parts which occupy the resources. The *GanttChart* shows the parts along the vertical axis and the elapsing time along the horizontal axis.

> **Note**
>
> The *GanttChart* shows the parts in the order in which they were processed by a resource for the first time. The part, which was processed by a resource first, is shown in the first lane in the chart. The part, which was processed by a resource after that, is shown in the lane below, etc.
>
> You cannot change the order of the parts.



With the default settings the *GanttChart* shows a resource for all parts with the same color.

The displayed resource states refer to the resource which is currently occupied by the displayed part.



Show Longer or Shorter Time Intervals

To show **longer time** intervals, hold down **Ctrl** and press the **-** key or roll the mouse wheel backward. This zooms the view in.

To show **shorter time** intervals, hold down **Ctrl** and press the **+** key or roll the mouse wheel forward. This zooms the view out.

To change the displayed time intervals in **greater steps**, hold down **Shift** in addition to **Ctrl**.

To return to the **default zoom factor**, hold down **Ctrl** and press the **0** key.

SimTalk

IsShown [SimTalk] - GanttChart

See also

setData [SimTalk]

getData [SimTalk]

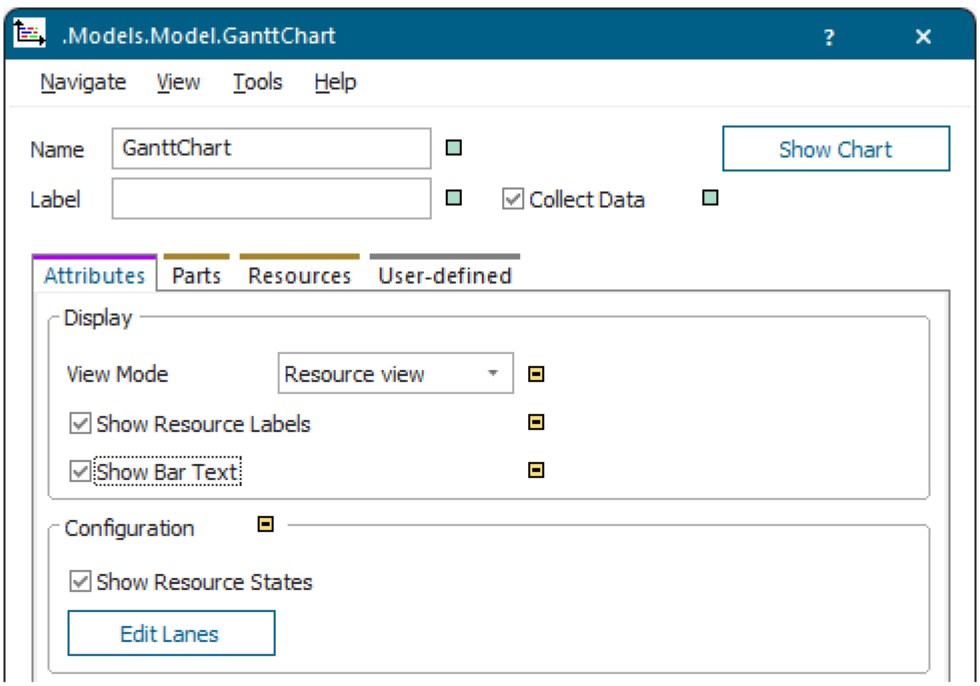
setLanes [SimTalk]

getLanes [SimTalk]

Tab Attributes

Tab Attributes [GanttChart]

Set how the *GanttChart* shows the data it collected on the tab **Attributes**.



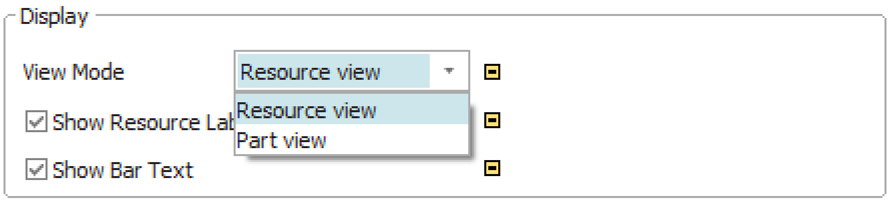
Remarks

You can:

- Select the View Mode, i.e., if the *GanttChart* displays data in the **Resource view** or the **Part view**.
- Select to Show Resource Labels [check box] or to hide them.
- Select to Show Bar Text [check box] or to hide it.
- Select to Show Resource States [check box] or to hide them.
- Edit Lanes on which the *Gantt data* is shown.

View Mode

Select the **View Mode** with which the *GanttChart* shows the *Gantt data*.



Remarks

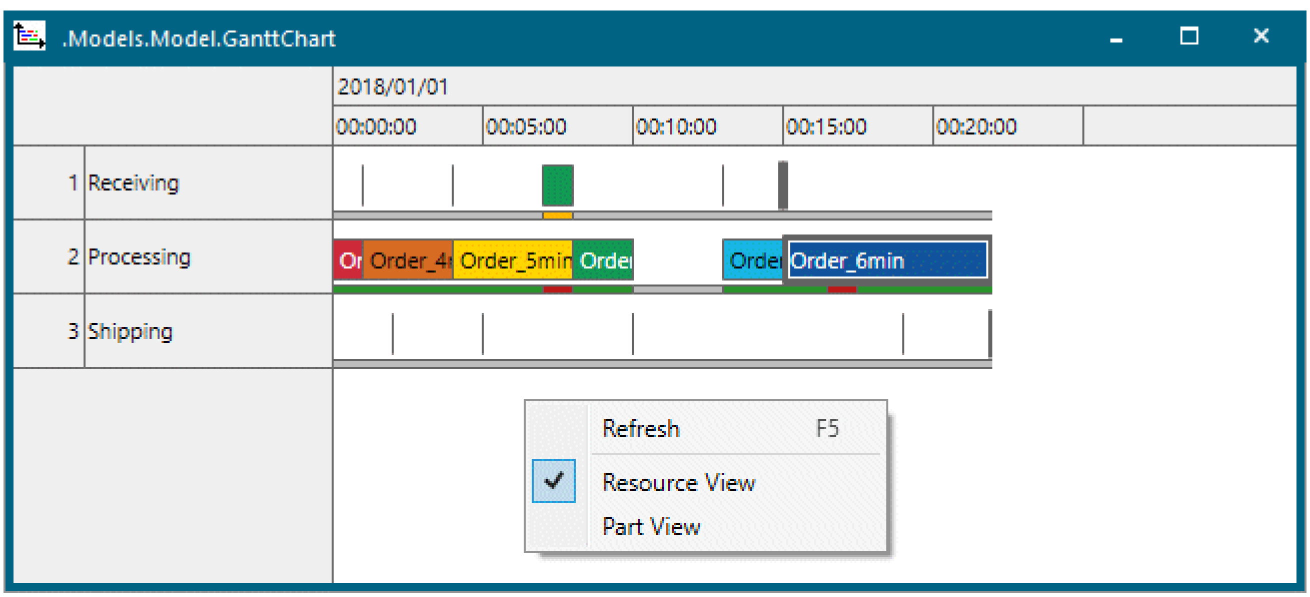
You can select one of these settings:

- **Resource View** shows the resources which are occupied by the parts. The *GanttChart* shows the resources along the vertical axis and the elapsing time along the horizontal axis.

Note

The *GanttChart* shows the resources in the order in which they processed parts for the first time. The resource, which processed a part first, is shown in the first lane in the chart. The resource, which processed a part after that, is shown in the lane below, etc.

You cannot change the order of the resources.

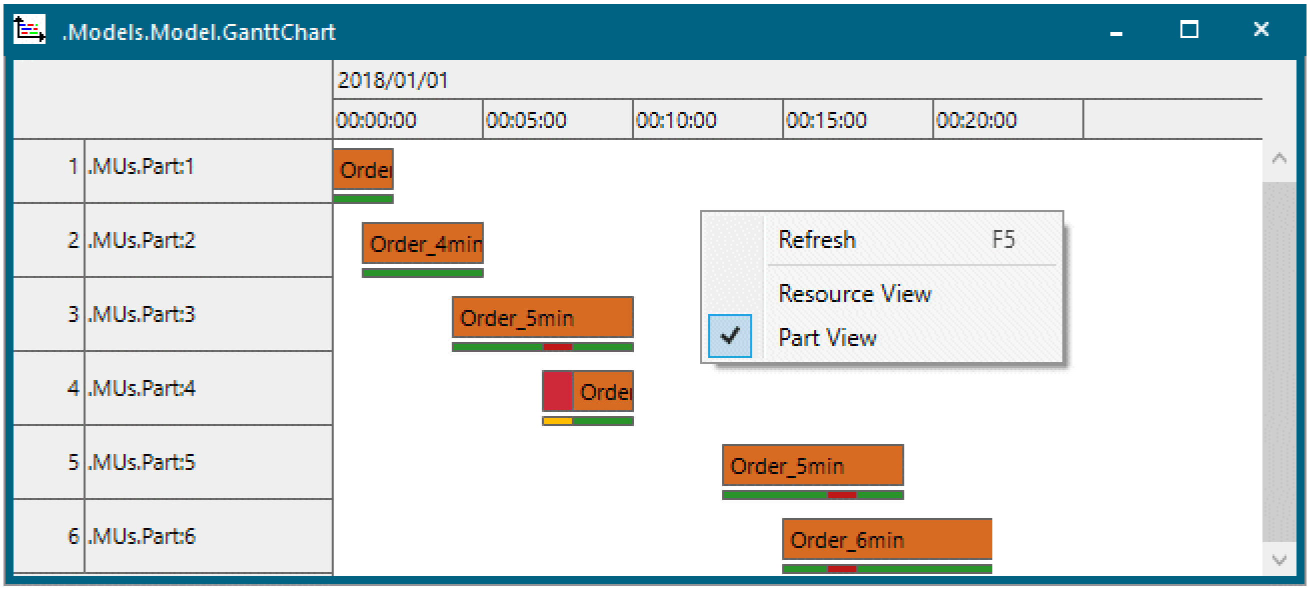


For parts being transported by a *Transporter* or carried by a *Worker*, **Resource View** shows the *Transporter* or the *Worker* as the resource instead of the *Track* or the *Footpath*.

With the default settings the *GanttChart* shows a part on all resources with the same color.

- **Part View** shows the parts which occupy the resources. The *GanttChart* shows the parts along the vertical axis and the elapsing time along the horizontal axis.

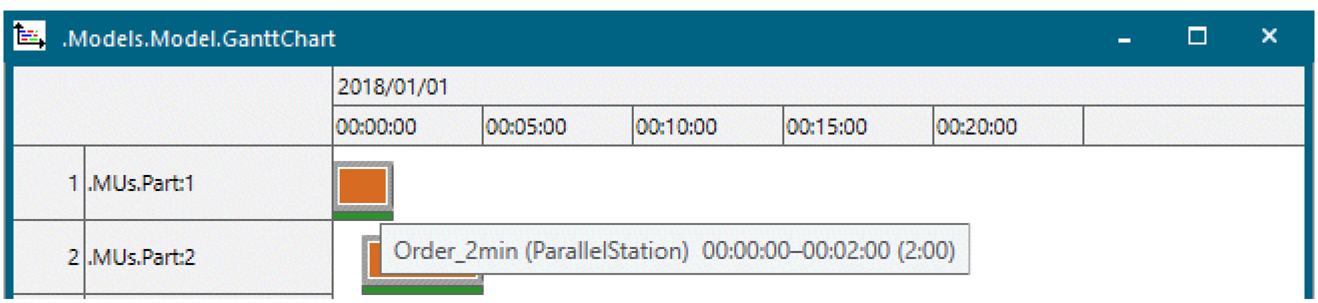
> **Note**
>
> The *GanttChart* shows the parts in the order in which they were processed by a resource for the first time. The part, which was processed by a resource first, is shown in the first lane in the chart. The part, which was processed by a resource after that, is shown in the lane below, etc.
>
> You cannot change the order of the parts.



With the default settings the *GanttChart* shows a resource for all parts with the same color.

You can select the respective **view mode** on the drop-down list or on the context menu.

The displayed resource states refer to the resource which is currently occupied by the displayed part.



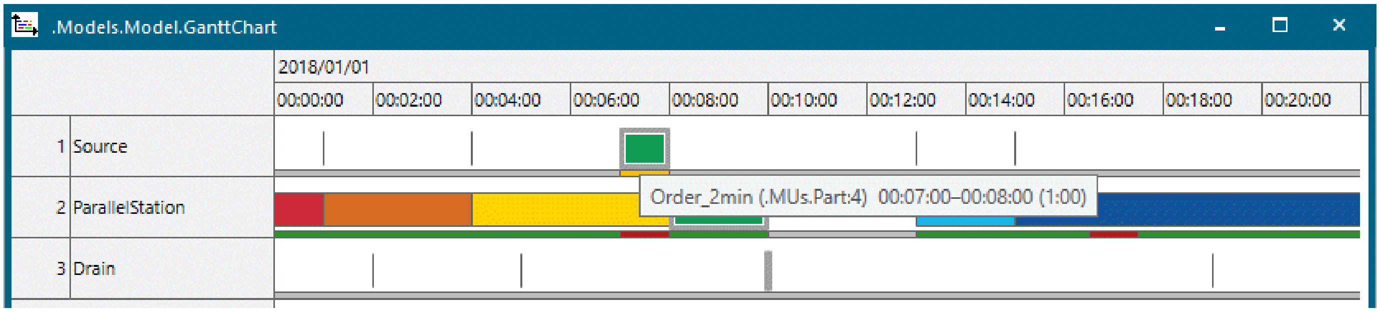
You can adjust the timescale of the *GanttChart* by holding down **Ctrl** while rolling the mouse wheel:

- Hold down **Ctrl** and roll the mouse forward to show shorter time intervals. This shows additional details of the displayed data.
- Hold down **Ctrl** and roll the mouse backward to show shorter time intervals. This shows a rough overview of the data.

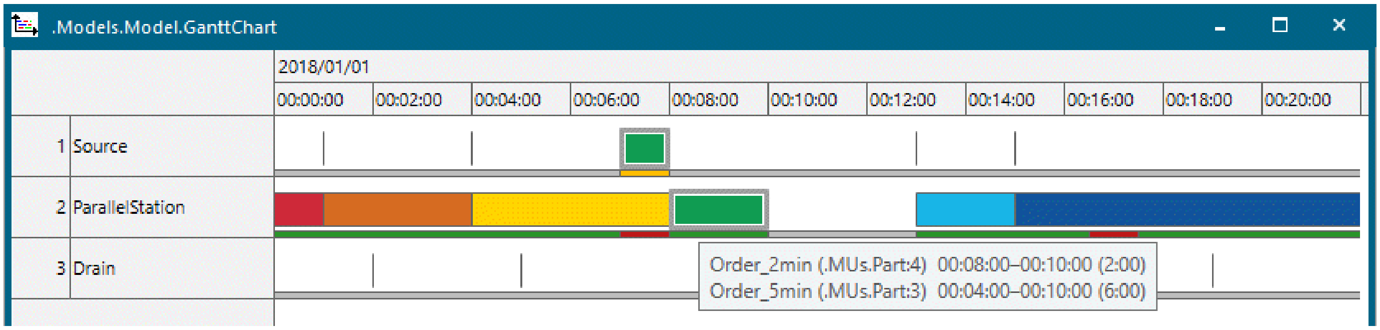
When you roll the mouse over a bar, **Resource View** highlights the respective part on all resources. **Part View** highlights the respective resource for all parts.

This way you can follow the way of the part across the simulation.

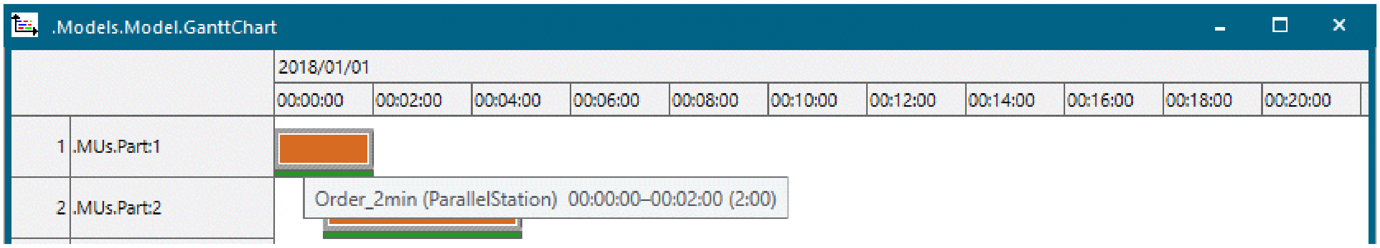
In **Resource View** the *Tooltip* shows the **name** of the part, its **absolute path**, the **start date**, the **end date**, and the **duration**.



If several parts are located on the same station, the *Tooltip* shows information about these parts. The part that moved last onto the station is located at the top of the list.



In **Part View** the *Tooltip* shows the **name** of the part, the **occupied resource**, the **start date**, the **end date**, and the **duration**.



Resource View refreshes the display of the *GanttChart* continuously during the simulation.

You have to manually refresh **Part View** by pressing **F5** or by selecting the context menu command **Refresh**.

To show the labels of the resource objects instead of their names in the column to the left, use **Show Resource Labels**.

To show or hide the Gantt bar text of the resources, use **Show Bar Text**. The Gantt bar texts can overlap.

To show or hide the resource states, use **Show Resource States**.

To set the **distance** of the Gantt bars **to the top** and the **height of the bars**, use **Edit Lanes**.

To set if the *GanttChart* shows the parts which occupy resources in **Part View** or the occupied resources in **Resource View**, use the attribute **ShowPartView**.

SimTalk

ShowPartView [SimTalk]

See also

setData [SimTalk]

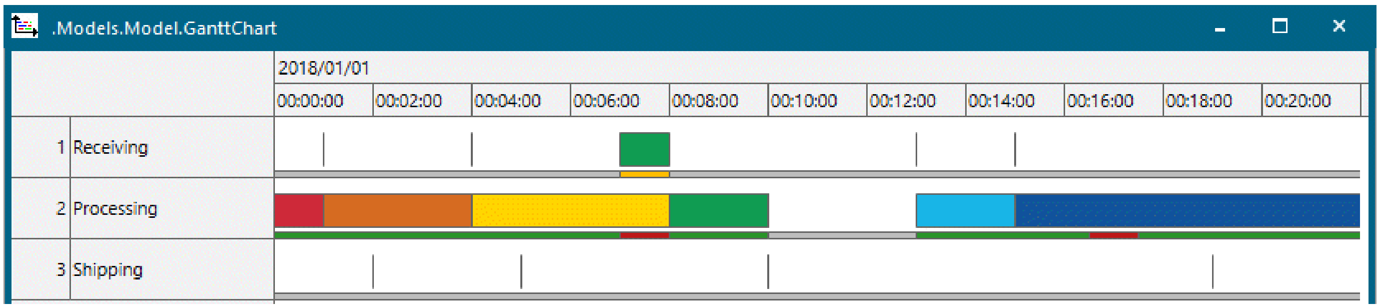
getData [SimTalk]

setLanes [SimTalk]

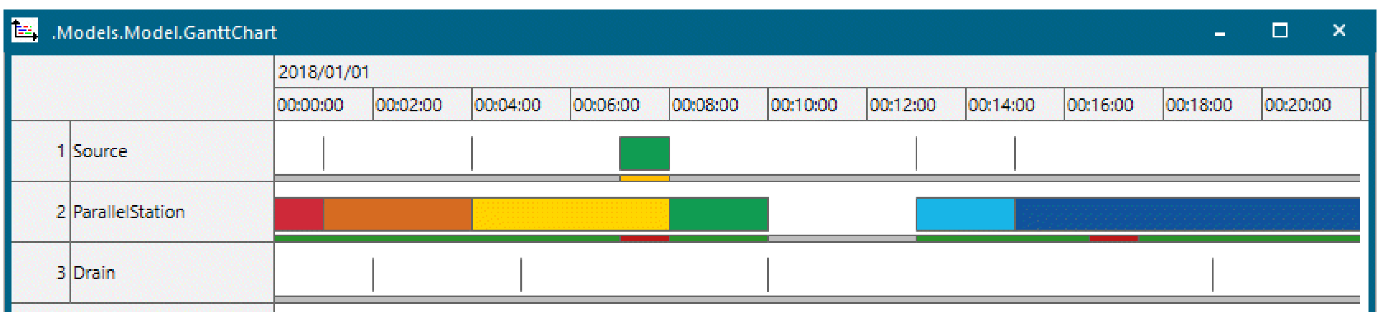
getLanes [SimTalk]

Show Resource Labels [check box]

To show the **labels of the resources**, select this check box.



To hide the **labels of the resources** and to show the **names of the resources** instead, clear the check box.



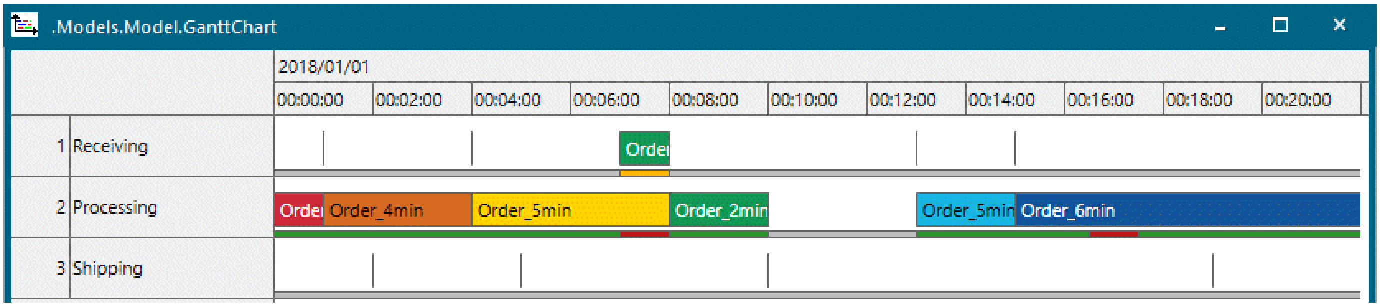
SimTalk

ShowResourceLabels [SimTalk]

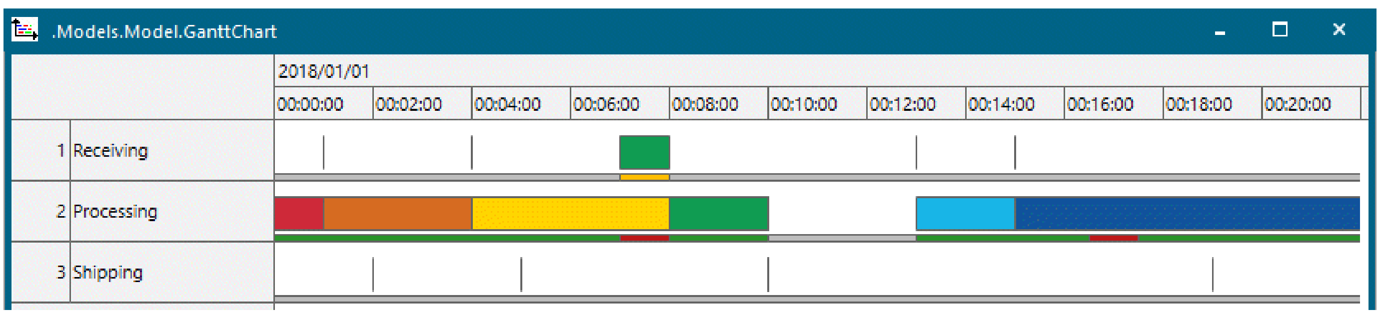
Show Bar Text [check box]

To show the **text on the Gantt bars**, select this check box.

Bar texts can overlap. The bar text shows the name of the part.



To hide the bar text of the resource, clear the check box.



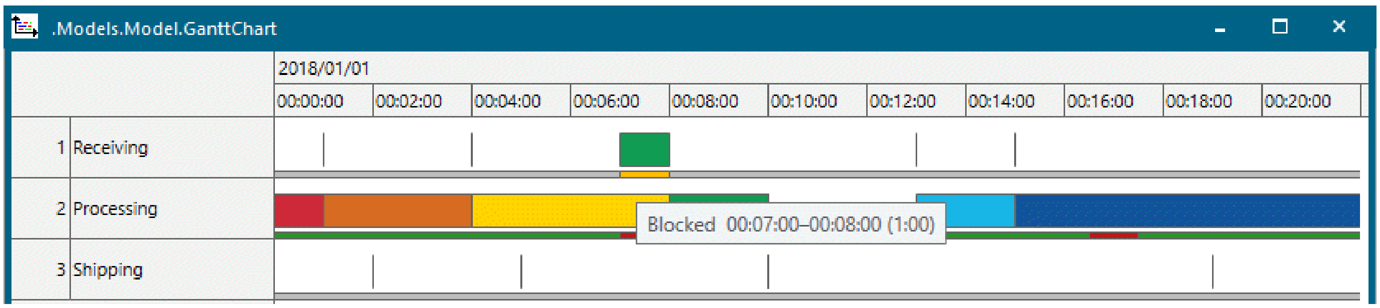
SimTalk

ShowBarText [SimTalk]

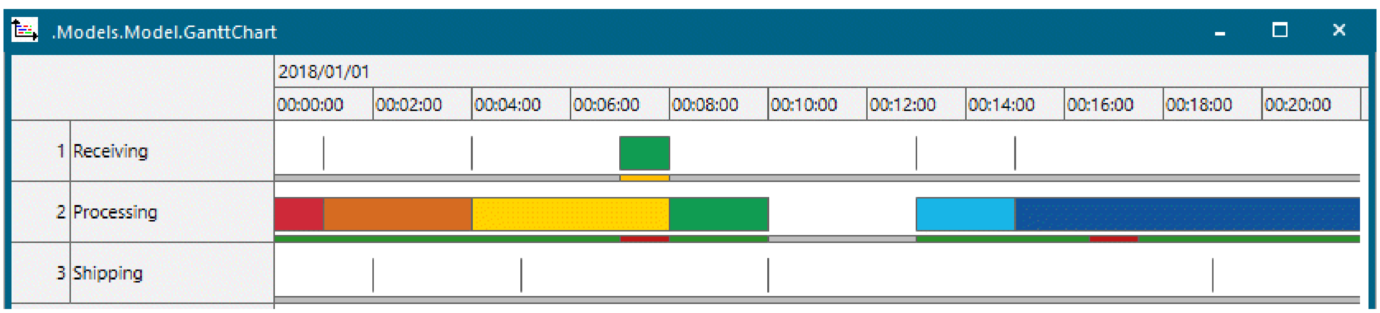
Show Resource States [check box]

To show the **states of the resources** as colored bars below the timeline, select this check box.

The colors of the parts are the same as the state colors of the objects, compare **States of the Material Flow Objects**.



To hide them, clear the check box.

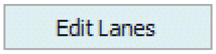


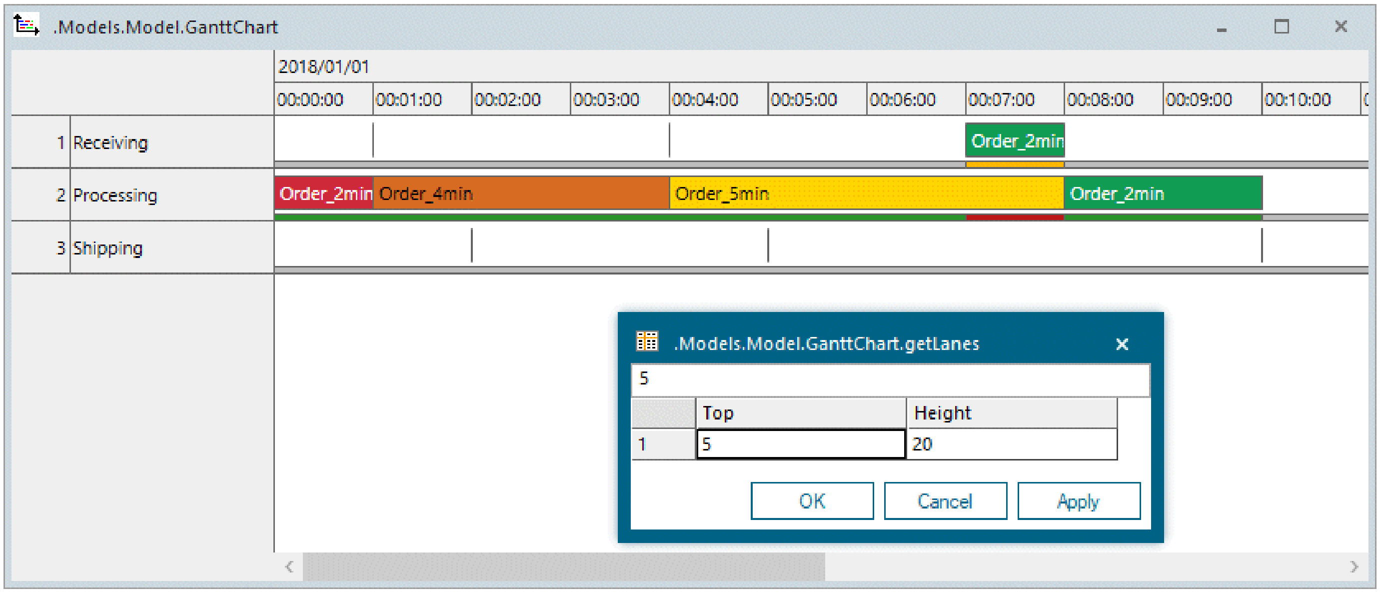
SimTalk

ShowResourceStates [SimTalk]

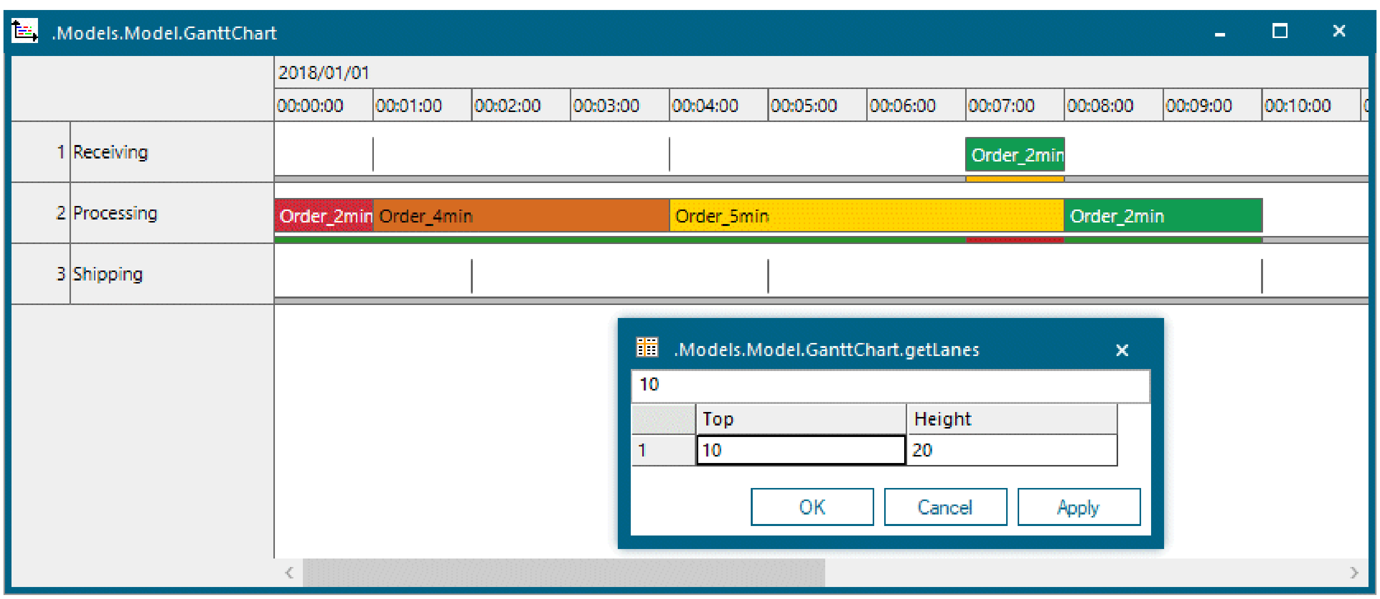
Edit Lanes

Click **Edit Lanes** and enter the **distance** of the Gantt bars **to the top** and the **height of the bars themselves**.





To apply the changed values, click **Apply** in the dialog **getLanes** and then **Apply** in the dialog of the *GanttChart*.



The *GanttChart* always collects data in lane 1. Specify your own data with the method **setData** to distribute them across several lanes.

SimTalk

setLanes [SimTalk]

getLanes [SimTalk]

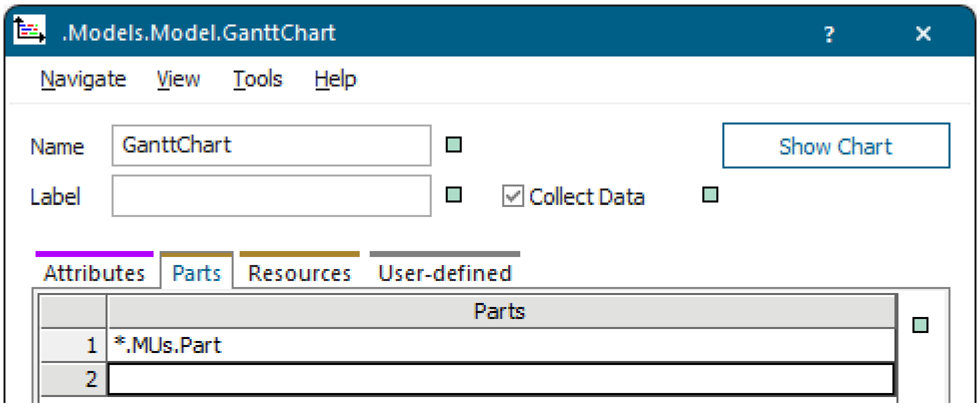
setData [SimTalk]

Tab Parts [GanttChart]

Add the parts that the *GanttChart* is to display on the tab **Parts**.

Remarks

First, clear the check box **Inheritance** so that it is looks like this . Then drag the part types, which you would like to watch, from the *Class Library* to an empty row on the tab.



Instead, you can also select the part in the *Class Library*, drag it on the icon of the *GanttChart*, and drop it there. This automatically deactivates inheritance and adds the part to an empty row on the tab **Parts**.

You can watch classes of parts or individual instances of parts.

You can edit the content of the text boxes with the commands on the **Context Menu of Embedded Lists**.

SimTalk

Parts [SimTalk]

Tab Resources [GanttChart]

Add the resources that the *GanttChart* is to display on the tab **Resources**.

Remarks

The *GanttChart* considers the **material flow objects** as resources, not the **resource objects**.

If you want to watch all resources in the simulation model, leave the tab **Resources** empty. The *GanttChart* then shows all resources on which parts, which you set on the tab **Parts**, move during the simulation.

First, clear the check box **Inheritance** so that it is looks like this . Then drag the resource types, which you would like to watch, from the *Class Library* to an empty row on the tab.

Instead, you can also select one or several resources in the *Frame*, drag it/them on the icon of the *GanttChart*, and drop it/them there. This automatically deactivates inheritance and adds the resources to an empty row each on the tab **Resources**.

You can watch resource classes or individual resource instances.

You can edit the content of the text boxes with the commands on the **Context Menu of Embedded Lists**.

SimTalk

Resources [SimTalk] - GanttChart

ShowResourceLabels [SimTalk]

ShowResourceLabels [SimTalk]

Tab User-defined

Define your own attributes as described under the **Tab User-defined**.

Navigate Menu

The commands are described under the **Navigate Menu**.

View Menu

The commands are described under the **View Menu**.

SimTalk

updateDialog [SimTalk]

Tools Menu

The **Tools Menu** provides commands to access its functions.

Edit Controls

Edit Observers

See also

Tools Menu [general description]

Help Menu

The commands are described under the **Help Menu**.

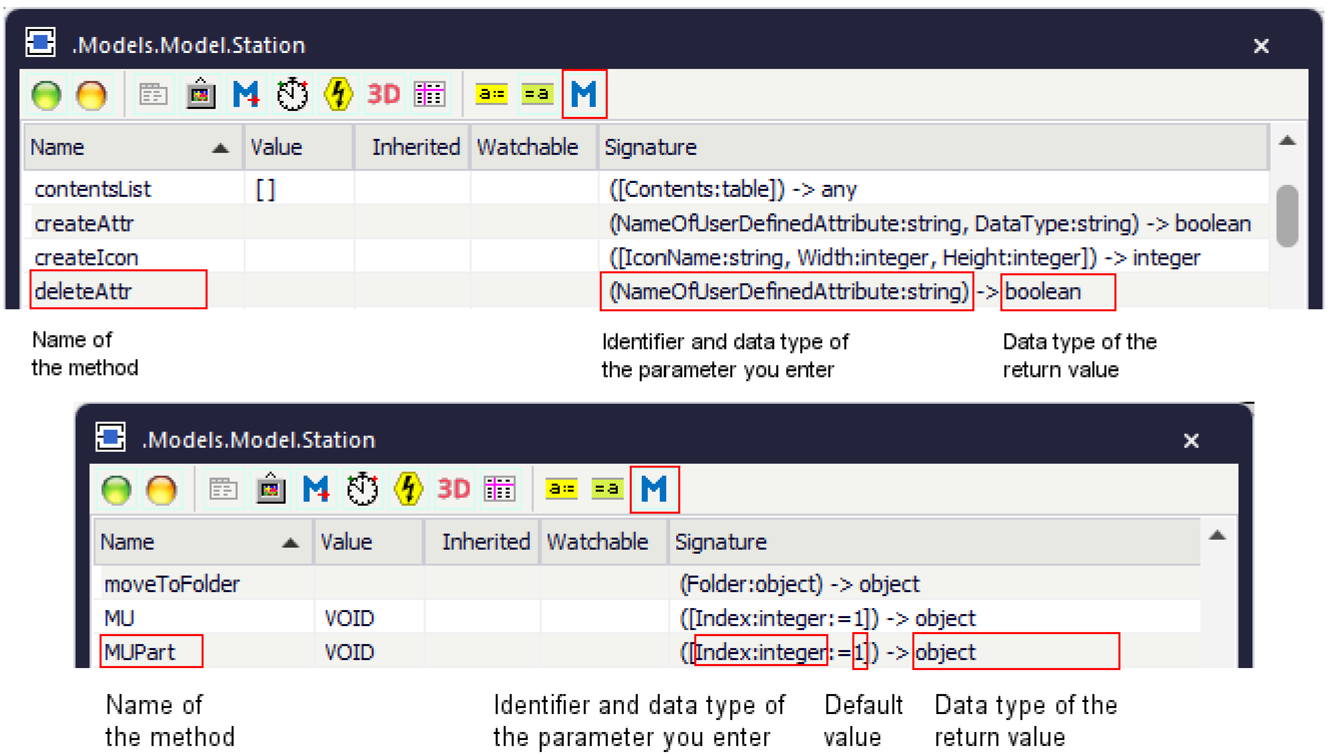
Methods of the GanttChart

_Methods of the GanttChart

The *GanttChart* provides:

- The *methods* listed in the table of contents to the left.
- The **Methods of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.